

Superstore Sales and Profit Analysis

A Data Analysis Project

by

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Overview:

This project explores the Superstore Sales Dataset to uncover key insights about sales performance, discount strategies, regional profitability and customer behavior and translate those findings into actionable business insights.

Objectives:

1: Sales and Profit Performance Analysis

Problem Statement: Analyze sales and profit performance across different product categories, customer segments, and regions to identify high-performing areas and areas needing improvement.

2: Discount Impact on Profitability

Problem Statement: Assess how discounting strategies affect profit margins and overall profitability across products and customer segments.

3: Regional and Geographic Performance

Problem Statement: Evaluate the performance of different regions and states to identify growth opportunities and operational inefficiencies.

4: Customer Profitability and Retention

Problem Statement: Identify the most valuable customers and analyze purchasing patterns to improve retention and marketing strategies.

Dataset Description:

Source: <https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>

Number of rows: 9994

Number of columns: 21

Column Info:

Row ID: Unique ID for each row.

Order ID: Unique Order ID for each Customer.

Order Date: Order Date of the product.

Ship Date: Shipping Date of the Product.

Ship Mode: Shipping Mode specified by the Customer.

Customer ID: Unique ID to identify each Customer.

Customer Name: Name of the Customer.

Segment: The segment where the Customer belongs.

Country: Country of residence of the Customer.

City: City of residence of the Customer.

State: State of residence of the Customer.

Postal Code: Postal Code of every Customer.

Region: Region where the Customer belongs.

Product ID: Unique ID of the Product.

Category: Category of the product ordered.

Sub-Category: Sub-Category of the product ordered.

Product Name: Name of the Product.

Sales: Sales of the Product.

Quantity: Quantity of the Product.

Discount: Discount provided.

Profit: Profit/Loss incurred.

Tools Used: Microsoft Excel, Power Query, MySQL, Power BI

Data Cleaning & Preparation:

Tool: Microsoft Excel

Steps:

Step 1: Loaded the .csv file into Excel through Power Query

Step 2: Fixed data types, removed errors, looked out for duplicate rows and their relevancy and added derived columns (year, month name, profit margin, shipping days).

The screenshot displays the 'Data Preview' tab in Microsoft Excel, showing a table with 22 rows and 6 columns. The columns are: Row ID, Order ID, Order Date, Ship Date, and Ship Mode. The data is filtered to show only 'Valid' rows. The 'Query Settings' pane on the right shows the query name 'Sample - Superstore' and a list of applied steps: Source, Promoted Headers, Trimmed Text, Cleaned Text, Filtered Rows, Added Custom, Removed Columns, Changed Type with Locale, Changed Type with Locale1, Changed Type, Removed Blank Rows, Trimmed Text1, Cleaned Text1, Capitalized Each Word, Added Custom1, Changed Type1, Added Custom2, Changed Type2, and Added Custom3.

Row ID	Order ID	Order Date	Ship Date	Ship Mode
1	CA-2016-152156	08/11/2016	11/11/2016	Second Class
2	CA-2016-152156	08/11/2016	11/11/2016	Second Class
3	CA-2016-138688	12/06/2016	16/06/2016	Second Class
4	US-2015-108966	11/10/2015	18/10/2015	Standard Class
5	US-2015-108966	11/10/2015	18/10/2015	Standard Class
6	CA-2014-115812	09/06/2014	14/06/2014	Standard Class
7	CA-2014-115812	09/06/2014	14/06/2014	Standard Class
8	CA-2014-115812	09/06/2014	14/06/2014	Standard Class
9	CA-2014-115812	09/06/2014	14/06/2014	Standard Class
10	CA-2014-115812	09/06/2014	14/06/2014	Standard Class
11	CA-2014-115812	09/06/2014	14/06/2014	Standard Class
12	CA-2014-115812	09/06/2014	14/06/2014	Standard Class
13	CA-2017-114412	15/04/2017	20/04/2017	Standard Class
14	CA-2016-161389	05/12/2016	10/12/2016	Standard Class
15	US-2015-118983	22/11/2015	26/11/2015	Standard Class
16	US-2015-118983	22/11/2015	26/11/2015	Standard Class
17	CA-2014-105893	11/11/2014	18/11/2014	Standard Class
18	CA-2014-167164	13/05/2014	15/05/2014	Second Class
19	CA-2014-143336	27/08/2014	01/09/2014	Second Class
20	CA-2014-143336	27/08/2014	01/09/2014	Second Class
21	CA-2014-143336	27/08/2014	01/09/2014	Second Class
22	CA-2015-127320	06/12/2015	12/12/2015	Standard Class

KPI List:

Objective 1: Sales and Profit Performance Analysis

Total Sales

Total Profit

Profit Margin (%)

Year-over-Year Growth (%)

Objective 2: Discount Impact on Profitability

Average Discount (%)

Profit Margin per Discount Band

Sales Volume vs Discount Correlation

Profit Lost due to Discounting

Objective 3: Regional and Geographic Performance

Sales by Region

Profit by Region

Profit Margin by Region (%)

Sales per Customer by Region

Top 5 States by Profit

Objective 4: Customer/Segment Profitability and Retention

Segment-Level Analysis

Total Customer

Total Sales

Total Profit

Profit Margin (%)

Average Sales per Customer

Customer-Level Analysis

Top Customers by Profit

Sales Frequency per Customer

Profit Margin per Customer

Objective 5: Category/Product Performance

Sales by Category

Profit by Category

Profit Margin by Category(%)

Sales per Customer by Category

Sales by Sub-Category

Profit by Sub-Category

Profit Margin by Sub-Category(%)

Sales per Customer by Sub-Category

Sales by Product

Profit by Product

Profit Margin by Product(%)

Sales per Customer by Product

Top 5 Products by Profit

Analysis & Findings:

Tool 1: MySQL

Objective 1: Sales and Profit Performance Analysis

KPI 1: Total Sales

```
79      # KPI 1: Total Sales
80
81 •    select sum(Sales) as Total_Sales
82      from sales_data;
```

Result Grid			Filter Rows:	Export: 	Wrap C
	Total_Sales				
▶	2297200.860299955				

KPI 2: Total Profit

```
84      # KPI 2: Total Profit
85
86 •    select sum(Profit) as Total_Profit
87      from sales_data;
```

Result Grid			Filter Rows:	Export: 	Wrap Cell Co
	Total_Profit				
▶	286397.0217000013				

KPI 3: Profit Margin (%)

```
91      # KPI 3: Profit Margin
92
93 •    select round(sum(Profit) / nullif(sum(Sales),0) * 100, 2) as Profit_Margin_Percentage
94      from sales_data;
```

Result Grid			Filter Rows:	Export: 	Wrap Cell Content: 
	Profit_Margin_Percentage				
▶	12.47				

KPI 4: Year-over-Year Growth (%)

```

97      # KPI 4: Year-over-Year Growth(%)
98      WITH sales_per_year AS (
99          SELECT
100             Year,
101             SUM(Sales) AS Total_Sales
102          FROM sales_data
103          GROUP BY Year
104          ORDER BY Year DESC
105      )
106      SELECT
107          Year,
108          Total_Sales,
109          ROUND(
110              (
111                  (Total_Sales - LAG(Total_Sales) OVER (ORDER BY Year))
112                  / NULLIF(LAG(Total_Sales) OVER (ORDER BY Year), 0)
113              ) * 100,
114              2) AS YOY_GROWTH
115      FROM sales_per_year
116      ORDER BY Year DESC;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Year	Total_Sales	YOY_GROWTH
▶	2017	733215.2551999999	20.36
	2016	609205.5980000008	29.47
	2015	470532.50899999985	-2.83
	2014	484247.4981000009	NULL

Objective 2: Discount Impact on Profitability

KPI 1: Average Discount (%)

```

122      # OBJECTIVE 2: Discount Impact on Profitability
123
124      # KPI 1: Average Discount (%)
125
126      SELECT round(avg(Discount) * 100, 2) as Avg_Discount_percentage
127      from sales_data;
128
129      # or weighted average discount
130
131      SELECT round((sum(Discount * Sales)/ round(sum(Sales),0)) * 100, 2) as Avg_Discount_percentage
132      from sales_data;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Avg_Discount_percentage
▶	15.62

KPI 2: Profit Margin per Discount Band

```

136  -- KPI 2: Profit Margin per Discount Band
137
138  select
139  case
140  when Discount >= 0 and Discount < 0.1 THEN '0-10%'
141  when Discount >= 0.1 and Discount < 0.2 THEN '10-20%'
142  when Discount >= 0.2 and Discount < 0.3 THEN '20-30%'
143  when Discount >= 0.3 and Discount < 0.4 THEN '30-40%'
144  when Discount >= 0.4 and Discount < 0.5 THEN '40-50%'
145  when Discount >= 0.5 and Discount < 0.6 THEN '50-60%'
146  when Discount >= 0.6 and Discount < 0.7 THEN '60-70%'
147  when Discount >= 0.7 and Discount < 0.8 THEN '70-80%'
148  when Discount >= 0.8 and Discount < 0.9 THEN '80-90%'
149  when Discount >= 0.9 and Discount <= 1.0 THEN '90-100%'
150  end as discount_band,
151  round(SUM(Sales),2) AS Total_Sales,
152  round(SUM(Profit),2) AS Total_Profit,
153  round((sum(Profit)/ nullif(sum(Sales), 0)) * 100 , 2) as profit_margin_percentage,
154  count(*) as rows_in_band
155  from sales_data
156  group by discount_band
157  order by discount_band;

```

discount_band	Total_Sales	Total_Profit	profit_margin_percentage	rows_in_band
0-10%	1087908.47	320987.6	29.51	4798
10-20%	81927.87	10448.17	12.75	146
20-30%	764594.37	90337.31	11.82	3657
30-40%	117720.11	-12760.42	-10.84	254
40-50%	121902.76	-25550.16	-20.96	217
50-60%	58918.54	-20506.43	-34.8	66
60-70%	6644.7	-5944.66	-89.46	138
70-80%	40620.28	-40075.36	-98.66	418
80-90%	16963.76	-30539.04	-180.03	300

KPI 3: Sales Volume vs Discount Correlation

To be explored visually

KPI 4: Profit Lost due to Discounting

```

163      # KPI 4: Profit Lost due to Discounting
164
165      select @margin0 := sum(profit)/nullif(sum(sales),0)
166      from sales_data
167      where discount = 0;

169      select
170      case
171      when discount >= 0 and discount < 0.1 then '0-10%'
172      when discount >= 0.1 and discount < 0.2 then '10-20%'
173      when Discount >= 0.2 and Discount < 0.3 THEN '20-30%'
174      when Discount >= 0.3 and Discount < 0.4 THEN '30-40%'
175      when Discount >= 0.4 and Discount < 0.5 THEN '40-50%'
176      when Discount >= 0.5 and Discount < 0.6 THEN '50-60%'
177      when Discount >= 0.6 and Discount < 0.7 THEN '60-70%'
178      when Discount >= 0.7 and Discount < 0.8 THEN '70-80%'
179      when Discount >= 0.8 and Discount < 0.9 THEN '80-90%'
180      when Discount >= 0.9 and Discount <= 1.0 THEN '90-100%'
181      end as discount_band,
182      round(sum(sales),2) as Total_Sales,
183      round(sum(profit),2) as Actual_Profit,
184      round((sum(Profit)/ nullif(sum(Sales), 0)) * 100 , 2) as Actual_Margin_Percentage,
185      round(@margin0 * 100, 2) as Baseline_Margin_Percentage,
186      round((sum(sales) * @margin0) - sum(profit), 2) as Estimated_Profit_Lost
187      from sales_data
188      group by discount_band
189      order by discount_band;

```

Result Grid						
Filter Rows:						
Export:						
Wrap Cell Content:						
	discount_band	Total_Sales	Actual_Profit	Actual_Margin_Percentage	Baseline_Margin_Percentage	Estimated_Profit_Lost
▶	0-10%	1087908.47	320987.6	29.51	29.51	0
	10-20%	81927.87	10448.17	12.75	29.51	13724.67
	20-30%	764594.37	90337.31	11.82	29.51	135256.41
	30-40%	117720.11	-12760.42	-10.84	29.51	47493.76
	40-50%	121902.76	-25550.16	-20.96	29.51	61517.59
	50-60%	58918.54	-20506.43	-34.8	29.51	37890.35
	60-70%	6644.7	-5944.66	-89.46	29.51	7905.18
	70-80%	40620.28	-40075.36	-98.66	29.51	52060.38

Objective 3: Regional and Geographic Performance

KPI 1: Sales by Region

```

197      # KPI 1: Sales by Region
198
199 •    select Region, round(sum(sales), 2) as total_sales
200      from sales_data
201      group by Region
202      order by total_sales;
203

```

Region	total_sales
South	391721.91
Central	501239.89
East	678781.24
West	725457.82

KPI 2: Profit by Region

```

204      # KPI 2: Profit by Region
205
206 •    select Region, round(sum(profit), 2) as total_profit
207      from sales_data
208      group by Region
209      order by total_profit;
210

```

Region	total_profit
Central	39706.36
South	46749.43
East	91522.78
West	108418.45

KPI 3: Profit Margin by Region (%)

```

212      # KPI 3: Profit Margin by Region (%)
213
214 •    select Region, round(sum(profit) / nullif(sum(sales), 0) * 100, 2) as profit_margin
215      from sales_data
216      group by Region
217      order by profit_margin;
218

```

Region	profit_margin
Central	7.92
South	11.93
East	13.48
West	14.94

KPI 4: Sales per Customer by Region

```
219 # KPI 4: Sales per Customer by Region
220
221 • select Region, round(sum(sales)/ nullif(count(distinct(Customer_ID)), 0), 2) as Sales_per_Customer
222 from sales_data
223 group by Region
224 order by Sales_per_Customer
225
```

Region	Sales_per_Customer
South	765.08
Central	796.88
East	1007.09
West	1057.52

KPI 5: Top 5 States by Profit

```
226 # KPI 5: Top 5 States by Profit
227
228 • select state, round(sum(profit), 2) as total_profit
229 from sales_data
230 group by state
231 order by total_profit desc
232 limit 5;
233
```

state	total_profit
California	76381.39
New York	74038.55
Washington	33402.65
Michigan	24463.19
Virginia	18597.95

Objective 4: Customer/Segment Profitability and Retention

Segment-Level Analysis

KPI 1: Total Customer

```
239      # KPI 1: Total Customer by segment
240
241 •   select Segment, count(distinct(Customer_ID)) as total_customers
242      from sales_data
243      group by Segment
244      order by total_customers desc;
245
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Segment	total_customers		
Consumer	409		
Corporate	236		
Home Office	148		

KPI 2: Total Sales

```
246      # KPI 2: Total Sales by segment
247
248 •   select Segment, round(sum(Sales), 2) as total_sales
249      from sales_data
250      group by Segment
251      order by total_sales desc;
252
```

Result Grid	Filter Rows:	Export:	Wrap Cell Cont
Segment	total_sales		
Consumer	1161401.34		
Corporate	706146.37		
Home Office	429653.15		

KPI 3: Total Profit

```

253      # KPI 3: Total Profit by Segment
254
255 •   select Segment, round(sum(Profit), 2) as total_profit
256      from sales_data
257      group by Segment
258      order by total_profit desc;
259

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Segment	total_profit		
Consumer	134119.21		
Corporate	91979.13		
Home Office	60298.68		

KPI 4: Profit Margin (%)

```

260      # KPI 4: Profit Margin (%) by Segment
261
262 •   select Segment, round(sum(Profit)/ nullif(sum(Sales), 0) * 100, 2) as profit_margin
263      from sales_data
264      group by Segment
265      order by profit_margin desc;
266

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Segment	profit_margin		
Home Office	14.03		
Corporate	13.03		
Consumer	11.55		

KPI 5: Average Sales per Customer

```

267      # KPI 5: Average Sales per Customer by Segment
268
269 •   select Segment, round(sum(Sales)/ nullif(count(distinct(Customer_ID)), 0), 2) as avg_sales_per_customer
270      from sales_data
271      group by Segment
272      order by avg_sales_per_customer desc;
273

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Segment	avg_sales_per_customer		
Corporate	2992.15		
Home Office	2903.06		
Consumer	2839.61		

Customer-Level Analysis

KPI 1: Top Customers by Profit

```

275 # Customer-Level Analysis
276
277 # KPI 1: Top Customers by Profit
278
279 • select Customer_ID, Customer_Name, Round(Sum(Profit), 2) as total_profit
280    from sales_data
281   group by Customer_ID, Customer_Name
282  order by total_profit desc
283  limit 7;
284

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
Customer_ID	Customer_Name	total_profit		
TC-20980	Tamara Chand	8981.32		
RB-19360	Raymond Buch	6976.1		
SC-20095	Sanjit Chand	5757.41		
HL-15040	Hunter Lopez	5622.43		
AB-10105	Adrian Barton	5444.81		
TA-21385	Tom Ashbrook	4703.79		
CM-12385	Christopher Martinez	3899.89		

KPI 2: Sales Frequency per Customer

```

285 # KPI 2: Sales Frequency per Customer
286
287 • select Customer_ID, Customer_Name, count(distinct Order_ID) as Order_Count
288    from sales_data
289   group by Customer_ID, Customer_Name
290  order by Order_Count desc
291  limit 7;
292

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
Customer_ID	Customer_Name	Order_Count		
EP-13915	Emily Phan	17		
EA-14035	Erin Ashbrook	13		
CK-12205	Chloris Kastensmidt	13		
SH-19975	Sally Hughsby	13		
NS-18640	Noel Staavos	13		
PG-18820	Patrick Gardner	13		
JE-15745	Joel Eaton	13		

KPI 3: Profit Margin per Customer

```

293      # KPI 3: Profit Margin per Customer
294
295 •   select Customer_ID, Customer_Name,
296      Round((Sum(Profit)/ nullif(Sum(Sales),0)) * 100, 2) as profit_margin
297   from sales_data
298  group by Customer_ID, Customer_Name
299  order by profit_margin desc
300  limit 20;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	Customer_ID	Customer_Name	profit_margin
▶	JC-15385	Jenna Caffey	47.53
	TC-20980	Tamara Chand	47.14
	RB-19360	Raymond Buch	46.15
	SR-20740	Steven Roelle	45.8
	BO-11425	Bobby Odegard	45.44
	HL-15040	Hunter Lopez	43.68
	AR-10540	Andy Reiter	43.65
	CM-12385	Christopher Martinez	43.55
	IG-15085	Ivan Gibson	43.04
	RS-19870	Roy Skaria	42.92
	NM-18445	Nathan Mautz	42.6
	NP-18685	Nora Pelletier	41.87
	SC-20095	Sanjit Chand	40.71
	MC-17425	Mark Cousins	40.67
	BF-11080	Bart Folk	40.64
	KC-16255	Karen Carlisle	39.89
	KW-16435	Katrina Willman	39.76
	HH-15010	Hilary Holden	39.66

Objective 5: Category/Product Performance

KPI 1: Sales by Category

```

305      # Objective 5: Category/Product Performance
306
307      # KPI 1: Sales by Category
308 •    select Category, round(sum(Sales), 2) as Total_Sales
309      from sales_data
310      group by Category
311      order by Total_Sales desc;
312

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content
Category	Total_Sales		
Technology	836154.03		
Furniture	741999.8		
Office Supplies	719047.03		

KPI 2: Profit by Category

```

313      # KPI 2: Profit by Category
314 •    select Category, round(sum(Profit), 2) as Total_Profit
315      from sales_data
316      group by Category
317      order by Total_Profit desc;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Category	Total_Profit		
Technology	836154.03		
Furniture	741999.8		
Office Supplies	719047.03		

KPI 3: Profit Margin by Category(%)

```

319      # KPI 3: Profit Margin by Category(%)
320 •    select Category, round((sum(Profit)/nullif(sum(Sales),0)) * 100, 2) as Total_Profit
321      from sales_data
322      group by Category
323      order by Total_Profit desc;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content: IA
Category	Total_Profit		
Technology	17.4		
Office Supplies	17.04		
Furniture	2.49		

KPI 4: Sales per Customer by Category

```

325      # KPI 4: Sales per Customer by Category
326 •    select Category, round(sum(Sales)/nullif(count(distinct(Customer_ID)), 0), 2) as sales_per_customer
327      from sales_data
328      group by Category
329      order by sales_per_customer;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Category	sales_per_customer		
Office Supplies	912.5		
Furniture	1049.5		
Technology	1217.11		

KPI 5: Sales by Sub-Category

```

334      # KPI 5: Sales by Sub-Category
335 •    select Sub_Category, round(sum(Sales), 2) as Total_Sale:
336      from sales_data
337      group by Sub_Category
338      order by Total_Sales desc;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Sub_Category	Total_Sales		
Phones	330007.05		
Chairs	328449.1		
Storage	223843.61		
Tables	206965.53		
Binders	203412.73		
Machines	189238.63		
Accessories	167380.32		
Copiers	149528.03		
Bookcases	114880		
Appliances	107532.16		
Furnishings	91705.16		
Paper	78479.21		
Supplies	46673.54		
Art	27118.79		
Envelopes	16476.4		
Labels	12486.31		
Fasteners	3024.28		

KPI 6: Profit by Sub-Category

```

340      # KPI 6: Profit by Sub-Category
341 •    select Sub_Category, round(sum(Profit), 2) as Total_Profit
342      from sales_data
343      group by Sub_Category
344      order by Total_Profit desc;

```

Result Grid		
	Filter Rows:	Export: Wrap Cell Content:
Sub_Category	Total_Profit	
Copiers	55617.82	
Phones	44515.73	
Accessories	41936.64	
Paper	34053.57	
Binders	30221.76	
Chairs	26590.17	
Storage	21278.83	
Appliances	18138.01	
Furnishings	13059.14	
Envelopes	6964.18	
Art	6527.79	
Labels	5546.25	
Machines	3384.76	
Fasteners	949.52	
Supplies	-1189.1	
Bookcases	-3472.56	
Tables	-17725.48	

KPI 7: Profit Margin by Sub-Category(%)

```

346      # KPI 7: Profit Margin by Sub-Category(%)
347 •    select Sub_Category, round((sum(Profit)/nullif(sum(Sales),0)) * 100, 2) as Total_Profit
348      from sales_data
349      group by Sub_Category
350      order by Total_Profit desc;

```

Result Grid		
	Filter Rows:	Export: Wrap Cell Content:
Sub_Category	Total_Profit	
Copiers	55617.82	
Phones	44515.73	
Accessories	41936.64	
Paper	34053.57	
Binders	30221.76	
Chairs	26590.17	
Storage	21278.83	
Appliances	18138.01	
Furnishings	13059.14	
Envelopes	6964.18	
Art	6527.79	
Labels	5546.25	
Machines	3384.76	
Fasteners	949.52	
Supplies	-1189.1	
Bookcases	-3472.56	
Tables	-17725.48	

KPI 8: Sales per Customer by Sub-Category

```

352      # KPI 8: Sales per Customer by Sub-Category
353 •   select Sub_Category, round(sum(Sales)/nullif(count(distinct(Customer_ID)), 0), 2) as sales_per_customer
354      from sales_data
355      group by Sub_Category
356      order by sales_per_customer;

```

Result Grid		
Filter Rows:		
Export: Wrap Cell Content:		
Sub_Category	sales_per_customer	
Fasteners	15.83	
Labels	44.44	
Art	54.9	
Envelopes	79.98	
Paper	128.44	
Furnishings	173.68	
Supplies	291.71	
Appliances	302.06	
Binders	312.94	
Accessories	353.12	
Storage	435.49	
Bookcases	589.13	
Phones	645.81	
Tables	792.97	
Chairs	807	
Machines	1911.5	
Copiers	2336.38	

KPI 9: Sales by Product

```

363      # KPI 9: Sales by Product
364 •   select Product_ID, Product_Name, round(sum(Sales), 2) as Total_Sale
365      from sales_data
366      group by Product_ID, Product_Name
367      order by Total_Sales desc;

```

Result Grid			
Filter Rows:			
Export: Wrap Cell Content: Fetch row			
Product_ID	Product_Name	Total_Sales	
TEC-CO-10004722	Canon Imageclass 2200 Advanced Copier	61599.82	
OFF-BI-10003527	Fellowes Pb500 Electric Punch Plastic Comb Binding Machine With M...	27453.38	
TEC-MA-10002412	Cisco Telepresence System Ex90 Videoconferencing Unit	22638.48	
FUR-CH-10002024	Hon 5400 Series Task Chairs For Big And Tall	21870.58	
OFF-BI-10001359	Gbc Docubind T1300 Electric Binding System	19823.48	
OFF-BI-10000545	Gbc Ibimaster 500 Manual Proclink Binding System	19024.5	
TEC-CO-10001449	Hewlett Packard Laserjet 3310 Copier	18839.69	
TEC-MA-10001127	Hp Designjet T520 Inkjet Large Format Printer - 24" Color	18374.9	
OFF-BI-10004995	Gbc Docubind P400 Electric Binding System	17965.07	
OFF-SU-10000151	High Speed Automatic Electric Letter Opener	17030.31	
TEC-MA-10000822	Lexmark Mx611Dhe Monochrome Laser Printer	16829.9	
OFF-SU-10002881	Martin Yale Chadless Opener Electric Letter Opener	16656.2	
OFF-BI-10001120	Ibico Epk-21 Electric Binding System	15875.92	
FUR-BO-10004834	Riverside Palais Royal Lawyers Bookcase, Royale Cherry Finish	15610.97	
TEC-MA-10001047	3D Systems Cube Printer, 2Nd Generation, Magenta	14299.89	
TEC-PH-10001459	Samsung Galaxy Mega 6.3	13943.67	
TEC-PH-10002885	Apple Iphone 5	12996.6	

KPI 10: Profit by Product

```

369      # KPI 10: Profit by Product
370 •    select Product_ID, Product_Name, round(sum(Profit), 2) as Total_Profit
371      from sales_data
372      group by Product_ID, Product_Name
373      order by Total_Profit desc;

```

Product_ID	Product_Name	Total_Profit
TEC-CO-10004722	Canon Imagedclass 2200 Advanced Copier	25199.93
OFF-BI-10003527	Fellowes Pb500 Electric Punch Plastic Comb Binding Machine With M...	7753.04
TEC-CO-10001449	Hewlett Packard Laserjet 3310 Copier	6983.88
TEC-CO-10003763	Canon Pc1060 Personal Laser Copier	4570.93
TEC-MA-10001127	Hp Designjet T520 Inkjet Large Format Printer - 24" Color	4094.98
TEC-MA-10003979	Ativa V4110Mdd Micro-Cut Shredder	3772.95
TEC-MA-10001047	3D Systems Cube Printer, 2Nd Generation, Magenta	3717.97
TEC-AC-10002049	Plantronics Savi W720 Multi-Device Wireless Headset System	3696.28
OFF-BI-10001120	Ibico Epk-21 Electric Binding System	3345.28
TEC-MA-10000045	Zebra Zm400 Thermal Label Printer	3343.54
OFF-AP-10002945	Honeywell Enviracaire Portable Hepa Air Cleaner For 17' X 22' Room	3247.02
TEC-CO-10002095	Hewlett Packard 610 Color Digital Copier / Printer	3124.94
TEC-AC-10003033	Plantronics Cs510 - Over-The-Head Monaural Wireless Headset Sy...	3085.03
TEC-CO-10001046	Canon Imagedclass D680 Copier / Fax	2799.96
OFF-BI-10003925	Fellowes Pb300 Plastic Comb Binding Machine	2518.06
OFF-BI-10004600	Ibico Ibimaster 300 Manual Binding System	2318.34
TEC-AC-10003870	Logitech Z-906 Speaker Sys - Home Theater - 5.1-Ch	2243.93

KPI 11: Profit Margin by Product(%)

```

375      # KPI 11: Profit Margin by Product(%)
376 •    select Product_ID, Product_Name, round((sum(Profit)/nullif(sum(Sales),0)) * 100, 2) as Total_Profit
377      from sales_data
378      group by Product_ID, Product_Name
379      order by Total_Profit desc;

```

Product_ID	Product_Name	Total_Profit
OFF-PA-10004082	Adams Telephone Message Book W/Frequently-Called Numbers Sp...	50
TEC-MA-10002927	Canon Imagedclass Mf7460 Monochrome Digital Laser Multifunction ...	50
OFF-PA-10004734	Southworth Structures Collection	50
OFF-PA-10004092	Tops Green Bar Computer Printout Paper	50
OFF-PA-10002499	Xerox 1890	50
OFF-LA-10003498	Avery 475	50
OFF-PA-10003845	Xerox 1987	49
OFF-EN-10001434	Strathmore #10 Envelopes, Ultimate White	49
OFF-PA-10004530	Personal Creations Ink Jet Cards And Labels	49
OFF-PA-10001450	Rediform S.O.S. Phone Message Books	49
OFF-LA-10004055	Color-Coded Legal Exhibit Labels	49
TEC-MA-10003979	Ativa V4110Mdd Micro-Cut Shredder	49
OFF-LA-10002034	Avery 478	49
OFF-PA-10002001	Xerox 1984	49
OFF-PA-10000919	Xerox 1918	49
OFF-PA-10003739	Xerox 1969	49
OFF-PA-10000565	Easy-Staple Paper	49

KPI 12: Sales per Customer by Product

```

381      #KPI 12: Sales per Customer by Product
382 •   select Product_ID, Product_Name,
383      round(sum(Sales)/nullif(count(distinct(Customer_ID)), 0), 2) as sales_per_customer
384      from sales_data
385      group by Product_ID, Product_Name
386      order by sales_per_customer;

```

Product_ID	Product_Name	sales_per_customer
OFF-AP-10002203	Eureka Disposable Bags For Sanitaire Vibra Groomer I Upright Vac	1.62
TEC-AC-10003433	Maxell 4.7Gb Dvd+R 5/Pack	2.34
OFF-AP-10002906	Hoover Replacement Belt For Commercial Guardsman Heavy-Duty ...	2.44
OFF-BI-10003784	Computer Printout Index Tabs	2.77
OFF-BI-10000494	Acco Economy Flexible Poly Round Ring Binder	2.87
OFF-FA-10000490	Oic Binder Clips, Mini, 1/4" Capacity, Black	3.12
OFF-BI-10004141	Insertable Tab Indexes For Data Binders	3.5
OFF-BI-10002813	Avery Reinforcements For Hole-Punch Pages	3.56
OFF-BI-10000831	Storex Flexible Poly Binders With Double Pockets	3.61
TEC-AC-10003709	Maxell 4.7Gb Dvd-R 5/Pack	3.8
OFF-AR-10001149	Avery Hi-Liter Comfort Grip Fluorescent Highlighter, Yellow Ink	3.9
OFF-FA-10001754	Stockwell Gold Paper Clips	4.05
OFF-AR-10004272	Newell 308	4.2
OFF-BI-10002012	Wilson Jones Easy Flow II Sheet Lifters	4.29
OFF-FA-10003059	Assorted Color Push Pins	4.3
OFF-BI-10000201	Avery Triangle Shaped Sheet Lifters, Black, 2/Pack	4.31
OFF-FA-10000840	Oic Thumb-Tacks	4.33

KPI 13: Top 5 Products by Profit

```

388      #KPI 13: Top 5 Products by Profit
389 •   select Product_ID, Product_Name, round(sum(Profit), 2) as total_profit
390      from sales_data
391      group by Product_ID, Product_Name
392      order by total_profit desc
393      limit 5;

```

Product_ID	Product_Name	total_profit
TEC-CO-10004722	Canon Imagedclass 2200 Advanced Copier	25199.93
OFF-BI-10003527	Fellowes Pb500 Electric Punch Plastic Comb Binding Machine With M...	7753.04
TEC-CO-10001449	Hewlett Packard Laserjet 3310 Copier	6983.88
TEC-CO-10003763	Canon Pc1060 Personal Laser Copier	4570.93
TEC-MA-10001127	Hp Designjet T520 Inkjet Large Format Printer - 24" Color	4094.98

Tool 2: Power BI

Pre-requisite Steps:

Step 1: Grouping Dataset Columns

Column Info:

Row ID: Unique ID for each row.

Order ID: Unique Order ID for each Customer. (Identifier)

Order Date: Order Date of the product. (Date/Time)

Ship Date: Shipping Date of the Product. (Date/Time)

Ship Mode: Shipping Mode specified by the Customer.

Customer ID: Unique ID to identify each Customer. (Identifier)

Customer Name: Name of the Customer. (Supporting Attribute)

Segment: The segment where the Customer belongs. (Category)

Country: Country of residence of the Customer. (Category)

City: City of residence of the Customer. (Category)

State: State of residence of the Customer. (Category)

Postal Code: Postal Code of every Customer. (Supporting Attribute)

Region: Region where the Customer belongs. (Category)

Product ID: Unique ID of the Product. (Identifier)

Category: Category of the product ordered. (Category)

Sub-Category: Sub-Category of the product ordered. (Category)

Product Name: Name of the Product. (Supporting Attribute)

Sales: Sales of the Product. (Numerical)

Quantity: Quantity of the Product. (Numerical)

Discount: Discount provided. (Numerical)

Profit: Profit/Loss incurred. (Numerical)

Step 2: STAR Schema Arrangement

Fact Table:

Row ID, Order ID, Order Date, Ship Date, Customer ID, Region, Product ID, Sales, Quantity, Discount, Profit

Dimension Tables:

Table 1: Dim_Date: Date, Year, Month, Month Name, Quarter

Link to Fact Table: Ship Date, Order Date

Table 2: Dim_Customer: Customer_ID, Customer Name, Segment

Link to Fact Table: Customer_ID

Table 3: Dim_Product: Product_ID, Product Name, Category, Sub-Category

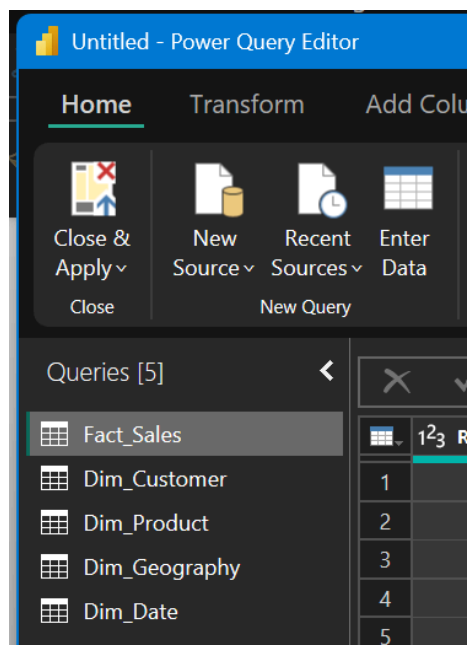
Link to Fact Table: Dim_Product

Table 4: Dim_Geography: Region, Country, City, State, Postal Code

Link to Fact Table: City (most granular)

Step 3: Data Transformation in Power Query

- Loaded the Excel dataset into Power Query Editor.
- Split the flat table into fact and dimension tables based on the star schema design.
- Retained relevant columns in each table for analysis.
- Applied basic data cleaning operations, including:
 - Removing duplicate rows
 - Ensuring consistent data types
 - Renaming columns for clarity

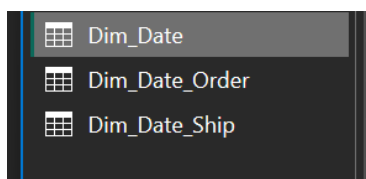


- Applied and loaded data from Power Query Editor
- Defined relationships between fact and dimension tables in Model View, linking foreign keys in the fact table to primary keys in the dimension tables.

Step 4: Building Relationships in Model View

Dim_Date Table:

- **Challenge:** Since we have both **Order Date** and **Ship Date** columns in the Dim_Date table, it causes overlapping dates, so when we try to relate Fact and Date on Order Date, Power BI cannot establish a many-to-one relationship.
- **Solution:** In order for proper relationships to be established between the dates in Dim_Dates table and the Fact_Sales table, I created role-playing date tables, keeping only one Date column in each respectively, and linked them properly with active/inactive relationships:



Challenge 2: Range Missing when marking as Date Table for YoY Growth.

Power BI wouldn't let me mark the above created Dim_Date_Order as a Date Table because it's missing some dates within the full order date range of my dataset, which violates the rule that a proper date dimension must have a continuous range of dates (no gaps).

Solution: Created a true continuous table

New Table in Model View

Entered the following DAX:

```
Dim_Date =
VAR MinDate = MIN('Fact_Sales'[Order_Date])
VAR MaxDate = MAX('Fact_Sales'[Order_Date])
RETURN
ADDCOLUMNS (
    CALENDAR (MinDate, MaxDate),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Month Name", FORMAT([Date], "MMMM"),
    "Quarter", "Q" & FORMAT(QUARTER([Date]), "0"),
    "Weekday Name", FORMAT([Date], "dddd")
)
```

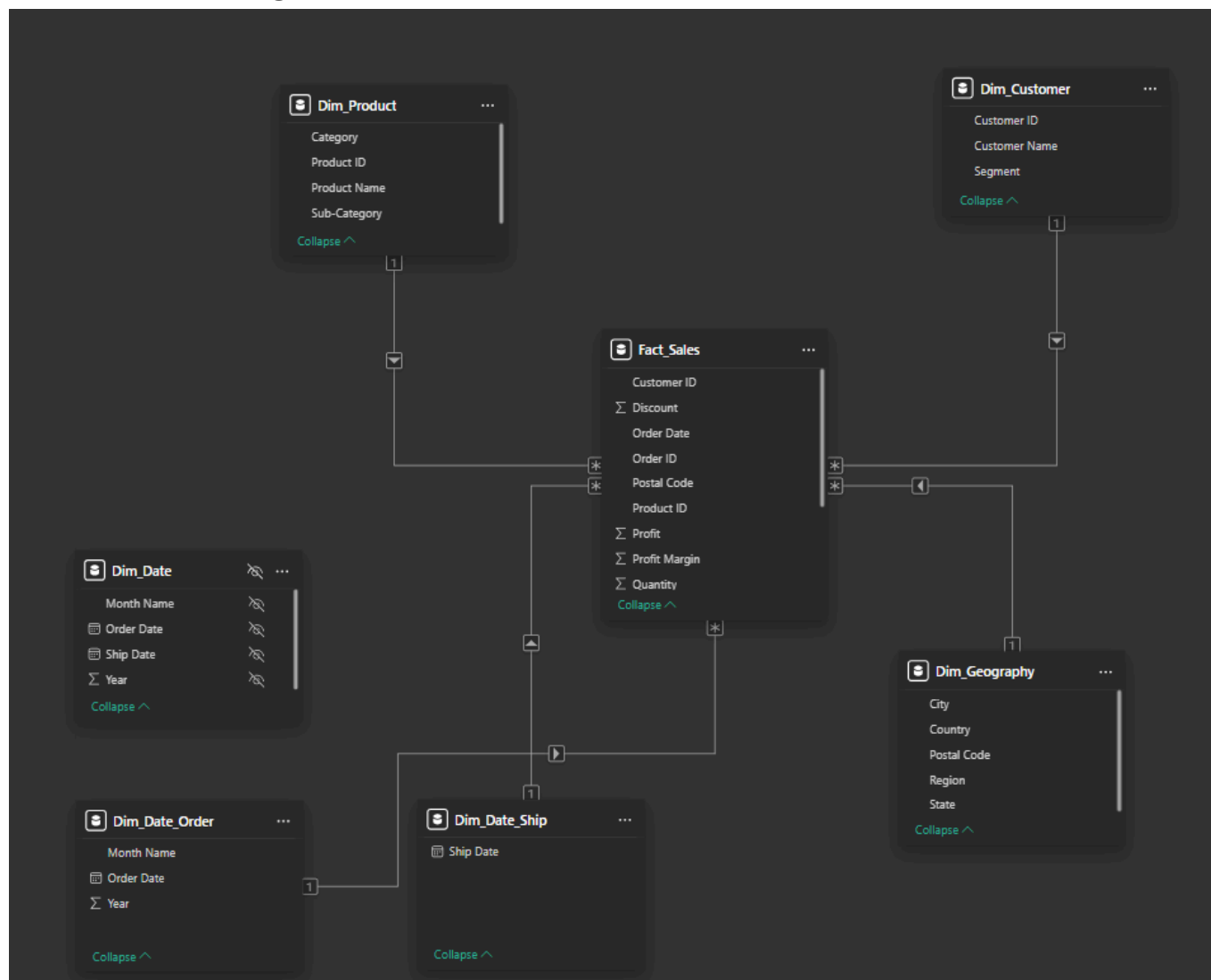
Right-click table and **mark as date table**

Dim_Geography Table:

- **Challenge:** Dim_Geography table contained two rows with the same Postal Code but different city names, causing duplicate key values. Since PostalCode was intended to be the unique key linking Dim_Geography to Fact Sales Table, this duplication prevented establishing a valid one-to-many relationship.

- **Solution:**
 - Grouped the **Dim_Geography** table by postal code to identify duplicate entries.
 - Created nested tables for each postal code group.
 - Added a custom column to merge multiple city names into one using Text.Combine(List.Distinct([Nest][City]), " / ").
 - Extracted the first row values for State, Region, and Country using [Nest]{0}[State], [Nest]{0}[Region], and [Nest]{0}[Country].
 - Removed the nested table column after extracting all required fields.
 - Verified that all Postal Code values were now unique.
 - Closed and applied changes to finalize the cleaned **Dim_Geography** table.
 - Established an active relationship between postal code in **Dim_Geography** and Fact Sales.

Final Data Modelling:



Step 5: Deciding Pages Layout:

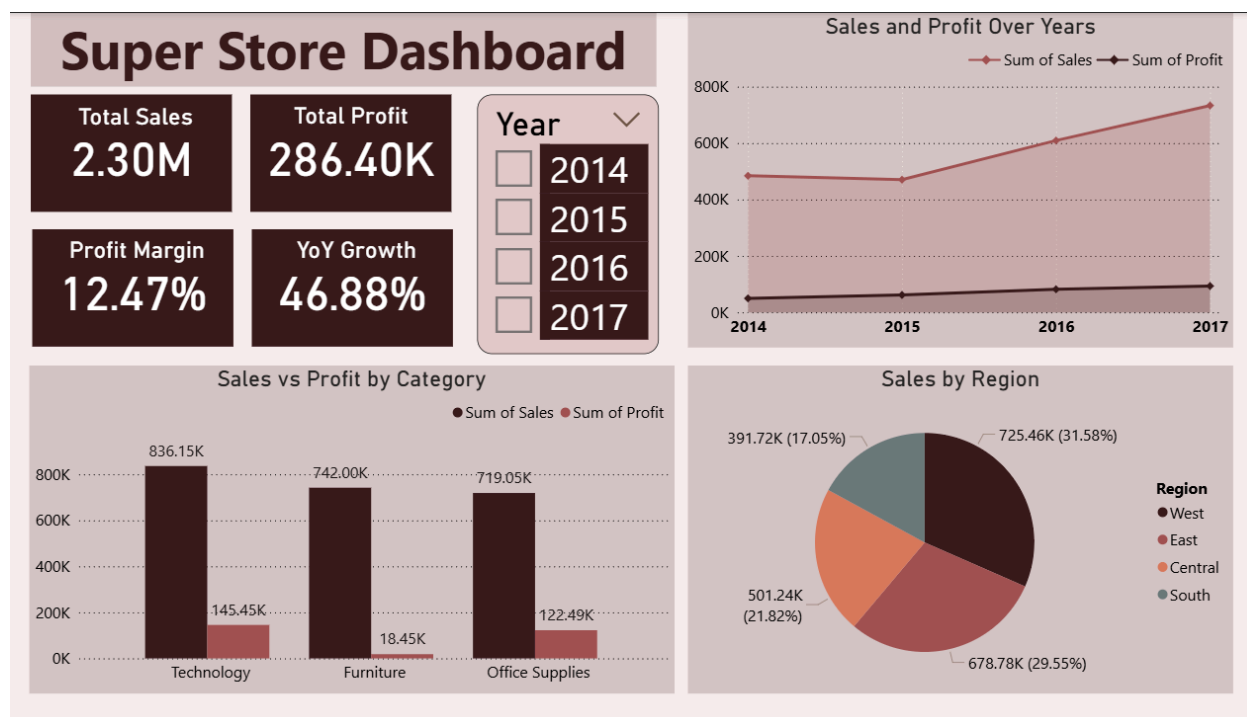
Page 1: Overview:

Attribute Selection:

Sales: (Numerical), Profit: (Numerical), Region, Category, Time (Year)

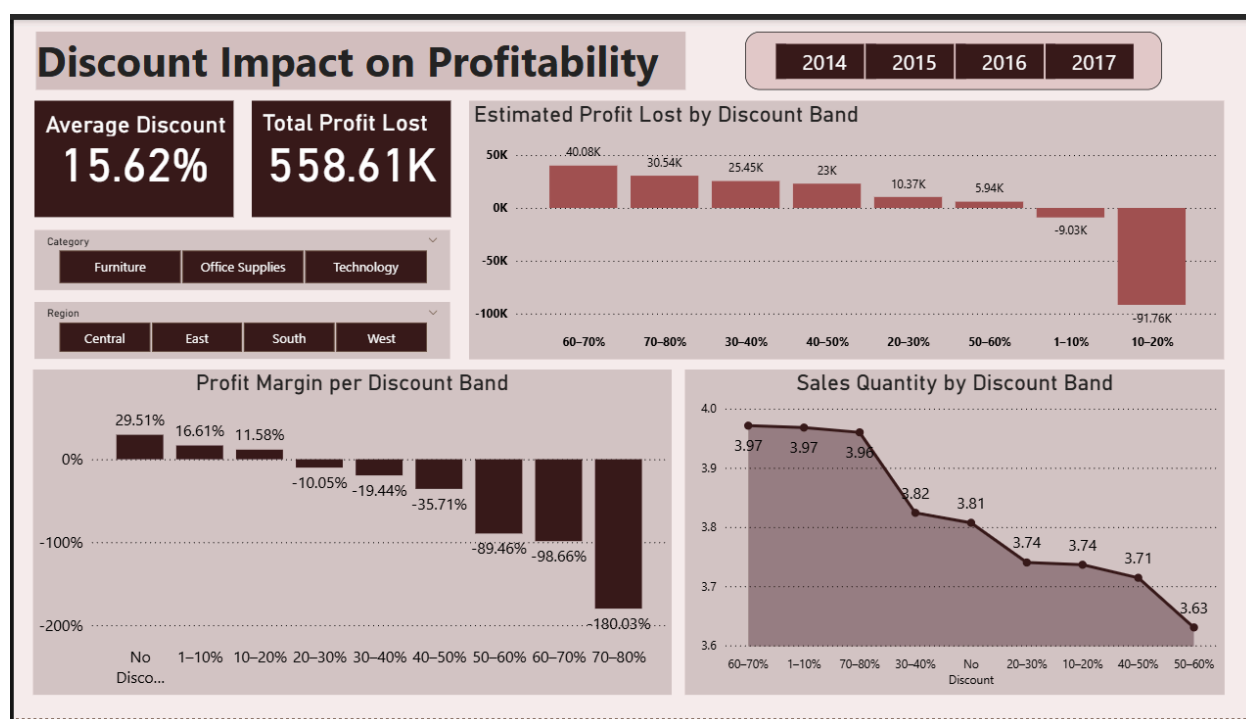
Visuals:

Total Sales
Total Profit
Profit Margin (%)
YoY Growth (%)
Sales vs Profit by Category
Sales by Region
Sales over Year/ Month



Page 2: Discount Impact on Profitability

- Average Discount (%)
- Total Profit Lost
- Profit Margin per Discount Band
- **Sales Volume vs Discount Correlation:** For this KPI, using a simple sum of 'Quantity' against each discount band would have given skewed results since the number of rows against each discount band would not be equal. For example, the no discount band has the most rows so it would naturally show more quantity. So I did **not** use raw sums. Instead, I normalised the quantity by using Average(Quantity).
- Profit Lost due to Discounting (per Discount Band)



Page 3: Regional and Geographic Performance

Sales by Region & Profit by Region

Profit Margin by Region (%)

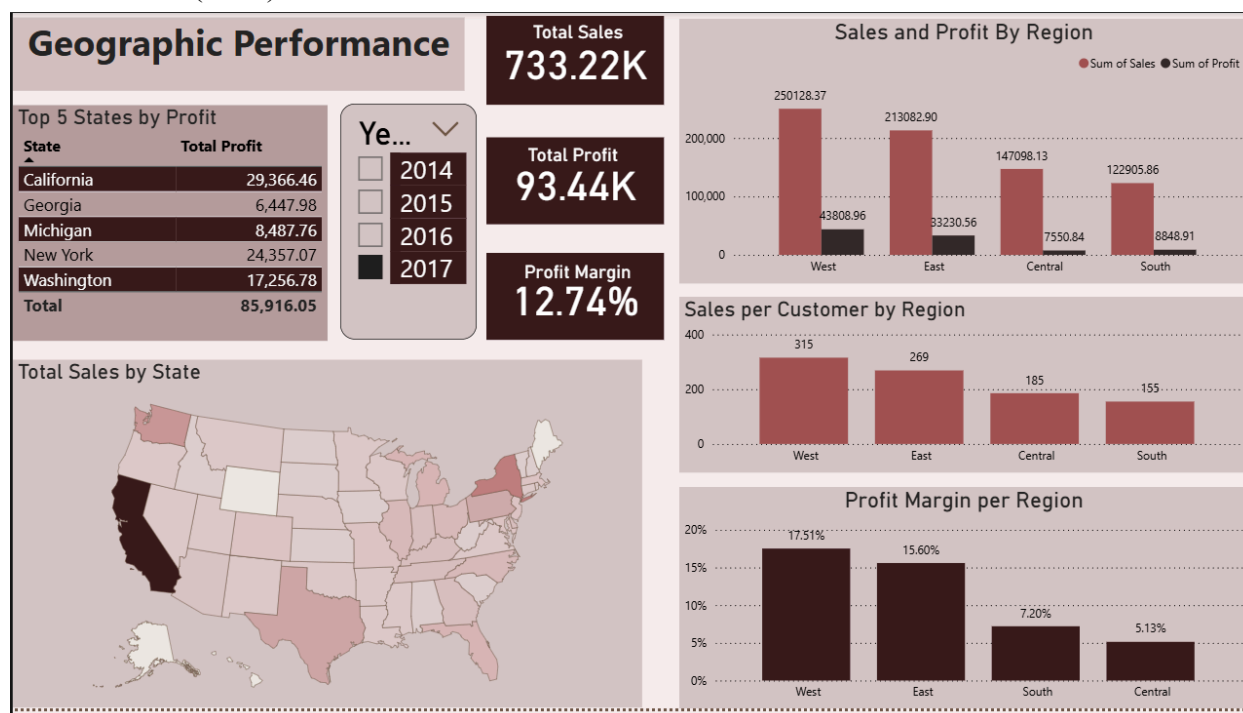
Sales per Customer by Region

Top 5 States by Profit

Total Sales by State (Map)

Basic KPI Cards (Total Sales, Total Profit, Profit Margin)

Slicer (Year)



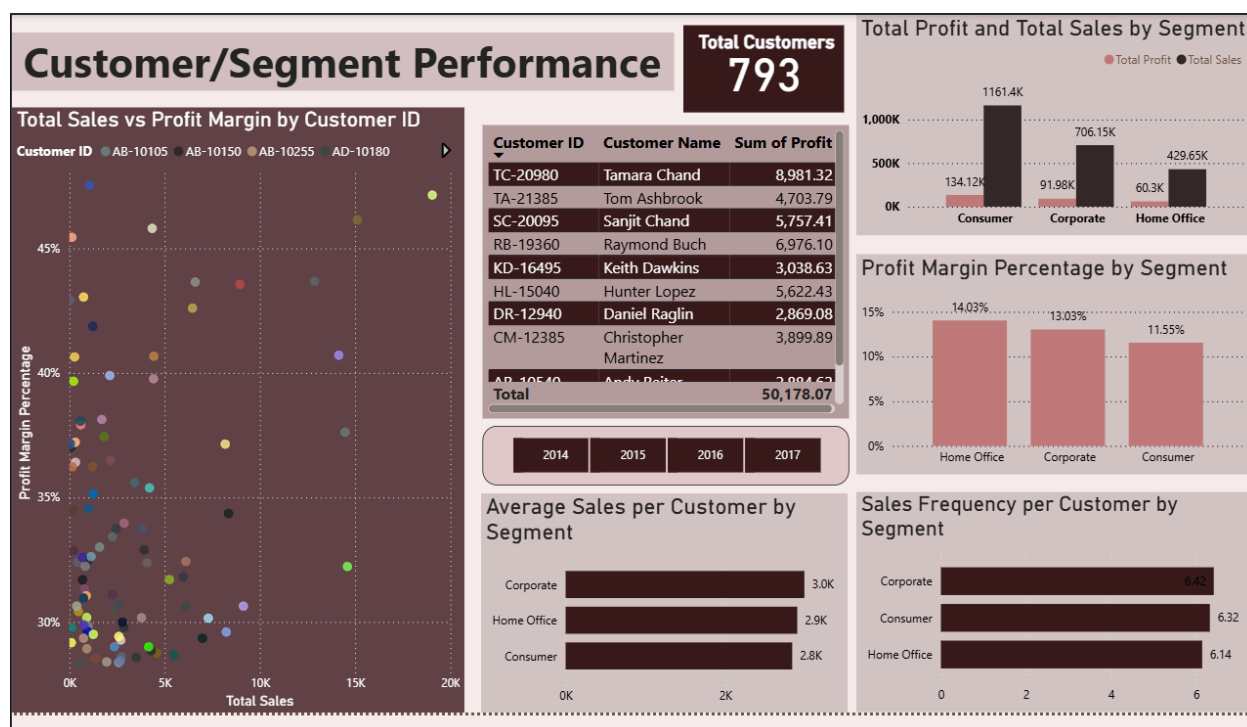
Page 4: Customer/Segment Profitability and Retention

Segment-Level Section

1. Average Sales per Customer by Segment (Horizontal Bar Chart)
2. Sales Frequency per Customer by Segment (Horizontal Bar Chart)
3. Profit Margin % by Segment (Column Chart)
4. Total Sales & Total Profit by Segment (Column Chart)

Customer-Level Section

6. Top 10 Customers by Profit (Table)
7. Profit Margin per Customer vs Total Sales (Scatter Plot)



Page 5: Category/Product Performance

Visuals:

Sales, Profit, Profit Margin % (KPI cards)

Sales vs Profit by Category (Bar chart)

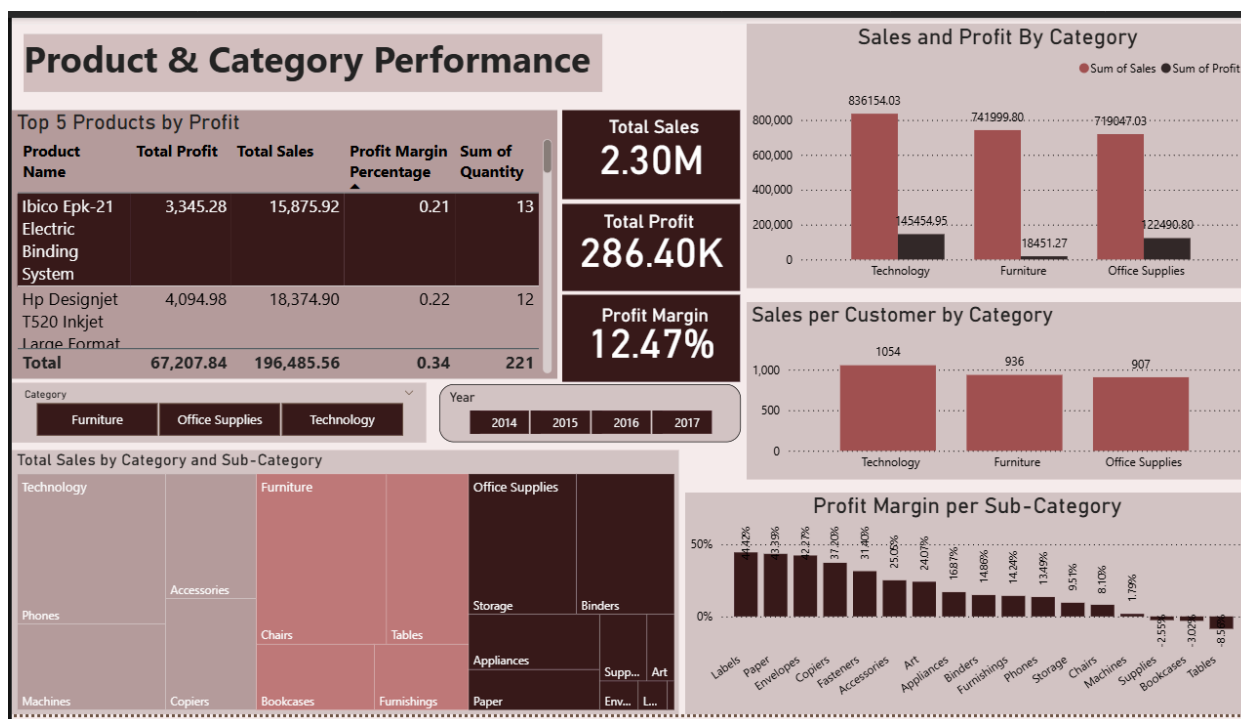
Sales per Customer by Category (Bar chart)

Sales by Category > Sub-Category (Treemap)

Profit Margin % by Sub-Category (Bar chart)

Top 10 Products by Profit (Table)

Slicers → Year, Category, Region



Comments on Slicers used:

Only those slicers that enhance insight without removing the core comparison dimension of the page were added. For example, Year and Category slicers allow to narrow the scope of analysis while still maintaining meaningful comparisons across regions, products, or customer segments. Conversely, slicers that duplicate the primary axis of analysis (such as Region on the Geography page or Category on the Category Performance page) were intentionally excluded, as they would eliminate the comparative context the visuals were built to showcase.

Observations made on Power BI pages:

Dashboard 1:

- 'Furniture' category has high sales but relatively very low profit across the years.
- 'South' region has underperformed over the years as compared to the remaining regions w.r.t Total Sales.

Dashboard 2:

- '50-60%' discount band has minimal profit loss.
- Profit Margin is only retained up until the 20% discount band. Discounts beyond 20% start to erode profit and above 50% are catastrophic when it comes to Profit Loss, and should only be considered in cases of clearance sale.

Dashboard 3:

- Western Region predominantly has the most Sales, Profit and Sales per Customer with California contributing the most, followed by the Eastern Region.

Dashboard 4:

- Consumer Segment contributes to the vast majority of Sales and Profit.

Dashboard 5:

- 'Technology' category brings in the highest amount of Sales and Profit throughout the years.
- 'Furniture' category has an alarmingly low profit margin with 'Bookcases' and 'Tables' leading to profit loss.

Insights, Analysis, Conclusions & Recommendations

Section 1: Dashboard-specific insights & analysis

Dashboard 1: Overview (Sales and Profit Performance)

Summary: The company faces a profitability quality problem masked by strong revenue growth. While sales have grown impressively (20-30% YoY), the 12.47% overall profit margin indicates that growth is not translating efficiently into bottom-line performance.

Insight 1: Furniture Category Performance Paradox

Finding: The Furniture category demonstrates high sales volume (\$741,200) but delivers alarmingly low profitability with only \$18,451 in profit, resulting in a profit margin of just 2.48%, significantly below the company average of 12.47%.

Context & Significance: This represents a critical business efficiency issue. While Furniture accounts for 32.3% of total sales (second-highest category), it contributes merely 6.4% of total profit. This means the company is dedicating substantial resources (inventory investment, warehouse space, shipping capacity, and sales effort) to a category that generates minimal return on investment.

Root Cause Analysis: Several interconnected factors likely drive this poor profitability:

- Cross-referencing with Dashboard 2 reveals that Furniture may be subject to aggressive discounting strategies that erode margins
- Bulky furniture items incur higher shipping and handling costs, which may not be adequately reflected in pricing
- The furniture wholesale market typically operates on thinner margins due to intense competition
- Product mix within furniture may include low-margin commodity items (tables, bookcases) rather than specialty or premium products

Business Implication: The company is essentially running a high-volume, low-margin operation in Furniture that ties up working capital without generating proportional profits. This creates opportunity cost: resources allocated to Furniture could potentially yield better returns if redirected to Technology or Office Supplies. Without intervention, continued growth in Furniture sales could paradoxically weaken overall company profitability.

Insight 2: South Region Underperformance

Finding: The South region consistently underperforms across all analyzed years, generating only \$391,721 in total sales (46% less than the West region (\$725,457) and 42% less than the East region (\$678,781)).

Context & Significance: Regional performance disparities of this magnitude suggest systemic issues rather than temporary market fluctuations. The South represents an underutilized market with significant growth potential, yet the company has failed to capture market share comparable to other regions.

Root Cause Analysis: Potential contributing factors include:

- Insufficient sales force presence or inadequate market penetration in Southern states
- Distribution and logistics challenges that may result in longer delivery times or higher shipping costs
- Weaker brand recognition or marketing investment in the South compared to other regions
- Competitive pressure from regional players who dominate Southern markets
- Demographic or economic differences that may require tailored product offerings or pricing strategies

Business Implication: The South region represents both a risk and an opportunity. As the lowest-performing region, it creates drag on overall company performance. However, this also means significant untapped potential: even bringing South to parity with Central region performance would add approximately \$110K in sales. This represents a strategic decision point: either invest aggressively to capture market share, or potentially rationalize operations in the region.

Insight 3: Strong Year-over-Year Growth Trajectory

Finding: The company demonstrates robust growth momentum with YoY increases of 23.88% (2015→2016), 29.55% (2016→2017), and 20.36% (2017→2018), indicating sustained market expansion and business scaling.

Context & Significance: Double-digit growth rates across consecutive years signal strong market demand, effective business execution, and successful scaling operations. This growth trajectory positions the company favorably within its competitive landscape and suggests that core business fundamentals are sound.

Root Cause Analysis: This sustained growth likely stems from:

- Successful customer acquisition and retention strategies
- Effective product portfolio management that aligns with market demand
- Strong performance in high-growth categories like Technology
- Possible market share gains from competitors
- Operational improvements that enable handling increased volume

Business Implication: The company is in a growth phase with positive momentum. This creates both opportunities and imperatives: opportunities to invest in market expansion, infrastructure, and talent; imperatives to ensure growth quality (profitability, not just revenue) and to address weak spots (like Furniture margins and South region performance) before they become more significant problems at scale.

Dashboard 2: Discount Impact on Profitability

Summary: \$156,285 in profit lost to discounting means the company is essentially giving away 54.6% of potential profits. This is not strategic discounting (targeted promotions, new customer acquisition); this is value destruction. The problem is systemic: high discounts (>30%) are normalized rather than exceptional.

Insight 4: Discount Threshold (20% Critical Limit)

Finding: Profit margins remain positive only up to the 20% discount band. Discounts exceeding 20% systematically erode profitability, with the 50-60% discount band

showing minimal profit loss only because it represents a small volume of transactions. Discounts above 50% are catastrophic, generating substantial profit loss.

Context & Significance: This finding reveals a clear mathematical relationship between discounting and profitability that should guide pricing policy. The company has lost \$156,285.28 in total profit due to discounting: representing 54.6% of actual realized profit. Put differently, without any discounting, the company's profit could have been \$442,682 instead of \$286,397.

Root Cause Analysis: The profit erosion at higher discount levels stems from:

- The average company profit margin is 12.47%, so discounts beyond this percentage guarantee losses
- High discounts (>30%) likely reflect desperation tactics: clearance of dead stock, competitive pressure, or poor inventory management
- The normalization of discounting may have trained customers to wait for sales rather than purchasing at full price
- Sales teams may be using discounts as an easy lever to close deals rather than demonstrating value

Business Implication: The current discounting strategy is destroying value. Every dollar of sales generated at >30% discount is actually reducing company profitability. This suggests three critical needs: (1) strict discount governance with approval thresholds, (2) alternative strategies for moving slow inventory (bundling, repositioning) rather than deep discounts, and (3) sales training focused on value-based selling rather than price-based closing.

Insight 5: Profit Lost Due to Discounting (The Hidden Cost)

Finding: The company has foregone \$156,285 in potential profit through discounting practices, with profit loss concentrated in discount bands above 30%.

Context & Significance: This "profit lost" represents more than half of the actual profit realized (\$286,397). Essentially, the company's profitability could increase by 54.6% if discounting strategies were optimized or eliminated. This is not theoretical: it represents real dollars that flowed to customers rather than to the bottom line.

Root Cause Analysis: The concentration of profit loss in high-discount bands suggests:

- Reactive rather than strategic discounting: using price cuts to solve problems (slow inventory, deal closure pressure) rather than as planned promotional tactics
- Lack of clear discount authority and approval processes
- Possible misalignment between sales incentives (revenue-focused) and company goals (profit-focused)
- Insufficient analysis of customer price sensitivity: high discounts may not be necessary to close many deals

Business Implication: Discount optimization represents one of the fastest paths to profitability improvement, requiring no additional sales, no new customers, and no market expansion - just better discipline and strategy around pricing. A 25% reduction in unnecessary discounting could add approximately \$39K to the bottom line, a 13.6% profit increase. This should be a top operational priority.

Dashboard 3: Regional and Geographic Performance

Summary: Over-reliance on Western region performance (especially California) creates vulnerability. The South's underperformance isn't just lost opportunity; it's a warning signal about the company's ability to replicate success in different markets.

Insight 6: Western Region Dominance (California Effect)

Finding: The Western region leads all performance metrics with \$725,457 in sales, \$108,418 in profit and the highest profit margin at 14.92%. California alone contributes the majority of Western region performance, followed by strong performance in the Eastern region.

Context & Significance: The West's outperformance is substantial: it generates 85% more sales than the South and 45% more than the Central region. This regional concentration creates both strengths and vulnerabilities: strengths in market penetration and operational efficiency in the West, but vulnerabilities from over-reliance on a single geographic market.

Root Cause Analysis: Western dominance likely reflects:

- Strong presence in California's large, affluent markets (tech industry concentration, population density)
- Mature distribution networks and established customer relationships in the West
- Possible proximity to warehousing or distribution centers that reduce shipping costs and improve service levels
- Cultural or demographic factors that align with the company's product offerings
- First-mover advantages or historical market development in Western states

Business Implication: While the West's strong performance should be maintained and leveraged, geographic concentration creates risk. Economic downturns, natural disasters, or increased competition in Western markets could disproportionately impact company performance. The company should view Western region success as both a strength to build on and a model to replicate in underdeveloped regions, while simultaneously working to diversify geographic revenue sources.

Insight 7: Sales per Customer Variation Across Regions

Finding: Sales per customer varies significantly by region, with the West showing the highest customer value and the South showing lower per-customer revenue generation.

Context & Significance: Per-customer sales metrics reveal not just market size differences but fundamental differences in customer behavior, product mix, or sales effectiveness across regions. High sales per customer in the West suggests either more affluent customer segments, better sales execution (upselling, cross-selling) or product mix skewed toward higher-value items.

Root Cause Analysis: Regional variations in sales per customer may stem from:

- Demographic and economic differences: Western markets may have higher household incomes or business spending capacity
- Differences in customer segment mix (more Corporate customers in some regions, more Consumers in others)
- Sales team effectiveness and training variations across regions
- Product availability or inventory allocation that favors certain regions
- Marketing effectiveness and brand strength varying by geography

Business Implication: Lower sales per customer in underperforming regions (South, Central) indicates opportunity for improvement through better sales execution, targeted marketing to higher-value customer segments, or product mix optimization. Even a 10% improvement in sales per customer in the South would generate approximately \$39K in additional revenue with minimal customer acquisition cost.

Dashboard 4: Customer/Segment Profitability and Retention

Summary: The wide variation in customer profitability (visible in the scatter plot) suggests the company is not systematically distinguishing between profitable and unprofitable customers. Some high-sales customers may be unprofitable due to high service costs or heavy discounting, while the company under-invests in truly profitable customer relationships.

Insight 8: Consumer Segment Dominance

Finding: The Consumer segment overwhelmingly drives business performance, contributing 51.48% of total sales (\$1,161,401) and 46.83% of total profit (\$129,185), despite having lower profit margins (11.71%) compared to Corporate (13.03%) and Home Office (13.83%) segments.

Context & Significance: Consumer segment dominance means the company's fortunes are closely tied to individual consumer spending patterns, economic conditions affecting households, and consumer-focused competitive dynamics. While this segment provides volume and scale, its lower margins mean the company must process more transactions to achieve profit targets.

Root Cause Analysis: Consumer segment characteristics driving this pattern include:

- Largest addressable market (consumers outnumber businesses)
- Higher transaction frequency but lower average order values
- Greater price sensitivity, leading to more discounting and lower margins
- Broader product category interest (consumers buy across all three categories)
- Possibly higher customer acquisition and service costs per dollar of revenue

Business Implication: The company should view its business model as fundamentally consumer-focused but should also recognize the efficiency opportunity in the Corporate and Home Office segments. These segments deliver comparable or better margins with potentially lower service costs (larger orders, more rational purchasing, less returns). A strategic initiative to grow Corporate and Home Office sales could improve overall profitability without abandoning the Consumer base.

Insight 9: Top Customer Concentration and Retention Risk

Finding: Analysis of top customers by profit reveals significant concentration, with the top 10 customers generating disproportionate profit contribution. The scatter plot of profit margin versus total sales shows high variability in customer profitability.

Context & Significance: Customer concentration creates both opportunity and risk. High-value customers represent the most efficient revenue source and should receive premium service and retention focus. However, over-reliance on a small number of customers exposes the company to significant risk if any of these relationships deteriorate or if competitors target these accounts.

Root Cause Analysis: High-value customer characteristics typically include:

- Larger organizations with substantial purchasing needs (Corporate segment)
- High purchase frequency indicating successful integration into their operations
- Diverse product category purchasing showing deep relationship
- Lower discount requirements suggesting strong value perception or relationship strength
- Potentially longer tenure indicating successful retention

Business Implication: The company should implement a formal key account management program for top-tier customers, including dedicated relationship managers, proactive service, customized solutions, and strategic pricing agreements that balance retention with profitability. Simultaneously, customer acquisition strategies should focus on identifying and attracting similar high-value profiles. The profit margin variation also suggests a need for customer profitability analysis (some high-sales customers may actually be unprofitable due to service costs or heavy discounting).

Dashboard 5: Category/Product Performance

Summary: The company generates most of its sales from three categories, but only Technology truly contributes healthy profits (17.35% margin). Office Supplies is adequate (16.98%), but Furniture (2.48%) is essentially a profitability black hole. As the business scales, if Furniture maintains its 32% revenue share, it will increasingly drag on overall margins.

Insight 10: Technology Category Leadership

Finding: The Technology category delivers the highest performance across all metrics: \$836,154 in sales (36.4% of total) and \$145,454 in profit (50.8% of total), with a healthy 17.35% profit margin. Technology consistently outperforms other categories year over year.

Context & Significance: Technology is the company's profit engine. Despite representing just over one-third of sales, it generates more than half of all profits. This indicates superior pricing power, better operational efficiency, or more favorable product economics in this category. Technology's strong performance essentially subsidizes the weak Furniture margins and enables overall business profitability.

Root Cause Analysis: Technology category strength likely derives from:

- Higher inherent margins in technology products due to brand premiums, rapid innovation cycles, and perceived value
- Less price sensitivity from customers who view technology purchases as investments rather than commodities
- Lower shipping and handling costs relative to product value (compact, high-value items)
- Strong demand drivers from business digitization trends and consumer technology adoption
- Possible category expertise and vendor relationships that yield favorable sourcing terms

Business Implication: Technology should be prioritized in strategic planning (marketing investment, inventory allocation, sales training, and potentially store/web placement).

The company should explore opportunities to expand its technology portfolio (new sub-categories, premium brands, accessories, services) as this represents the clearest path to profitable growth. Success in Technology can fund strategic investments needed in other areas.

Insight 11: Furniture Sub-Category Profit Losses (Bookcases and Tables)

Finding: Within the already-struggling Furniture category, specific sub-categories are particularly problematic. Bookcases and Tables are generating profit losses, dragging down the entire Furniture category performance and contributing to the category's 2.48% profit margin.

Context & Significance: This granular finding is critical: it means Furniture's poor performance is not uniformly distributed but concentrated in specific sub-categories. The company is not just earning low margins on Bookcases and Tables; it's actually losing money on every sale. These products are subsidized by better-performing furniture items and by other categories entirely.

Root Cause Analysis: Sub-category profit losses typically indicate:

- Pricing below cost due to competitive pressure or poor price management
- High return rates or quality issues creating hidden costs
- Excessive discounting to move slow-selling inventory
- Unfavorable shipping economics (large, heavy items with high damage rates)
- Possible sourcing issues where wholesale costs are too high relative to achievable retail prices
- Product obsolescence or style changes leaving the company with unsellable inventory

Business Implication: Bookcases and Tables require immediate action: either fix or exit. Options include: (1) price increases to restore profitability, accepting volume loss; (2) vendor renegotiation to reduce wholesale costs; (3) product repositioning toward higher-margin designs or materials; (4) discontinuation and inventory clearance, exiting these sub-categories entirely. Continuing to sell money-losing products makes no business sense unless they serve a strategic purpose (traffic drivers, category completeness) that generates compensating revenue elsewhere. A detailed sub-category P&L analysis should be the immediate next step.

SECTION 3: CONCLUSIONS

Primary Conclusions

1. **Technology is the Profit Engine:** With 17.35% margins and 50.8% of total profit from 36.4% of sales, Technology subsidizes the rest of the business. Strategic priority should be Technology category expansion and optimization.
2. **Furniture Requires Urgent Intervention:** At 2.48% profit margin with sub-categories (Bookcases, Tables) generating losses, Furniture is destroying value. The company must either fix or exit this category: status quo is not viable.
3. **Discounting is Out of Control:** Losing 54.6% of potential profit (\$156K) to discounting indicates lack of discipline and strategy. Discount governance and sales training must become immediate priorities.
4. **Geographic Imbalance Creates Risk:** West region dominance (especially California) and South region underperformance create concentration risk and untapped opportunity simultaneously.
5. **Customer Profitability Requires Segmentation:** Wide variation in customer profitability suggests need for explicit customer tier management: premium service for high-value customers, efficiency focus for marginal accounts.
6. **Growth Quality Trumps Growth Quantity:** 20-30% YoY growth is impressive but not sustainable if margins continue to erode. The next phase must prioritize profitable growth over pure volume expansion.

Secondary Conclusions

7. **Corporate and Home Office Segments are Underutilized:** These segments deliver better margins (13.03%, 13.83%) than Consumer (11.71%) but represent only 48.5% of sales. Segment mix optimization could improve overall profitability without reducing total revenue.

8. **Regional Performance Gaps Indicate Execution Issues:** The magnitude of difference between West and South performance (85% sales gap) cannot be explained by market size alone; this reflects execution capability gaps in sales, distribution, or marketing.
9. **Product Portfolio Needs Pruning:** Loss-making sub-categories and low-margin products create complexity and costs without contributing to profits. A rigorous SKU rationalization could improve margins and operational efficiency.
10. **Operational Scaling Has Been Successful:** The company's ability to grow 20-30% YoY while maintaining (not improving, but also not collapsing) margins suggests effective operational execution. This is a strength to build upon.

SECTION 4: RECOMMENDATIONS

Based on the analysis findings, the following recommendations are proposed to address identified profitability and performance issues:

1. Establish Discount Management Controls

The analysis reveals that \$558,610 in potential profit has been lost to discounting, with discounts above 20% consistently eroding profitability. Implementing formal discount approval thresholds and guidelines could help preserve margins while maintaining competitive pricing. Sales teams would benefit from training focused on value-based selling rather than relying primarily on discounting to close deals.

2. Re-evaluate Furniture Category Strategy

With only a 2.48% profit margin and specific sub-categories (Bookcases, Tables) generating losses, the Furniture category requires immediate strategic review. Options to consider include renegotiating supplier terms to improve costs, adjusting pricing strategies, or potentially discontinuing the lowest-performing sub-categories. A detailed sub-category profitability analysis would help determine which Furniture products are worth maintaining in the portfolio.

3. Invest in South Region Growth

The South region significantly underperforms other regions (\$391K in sales vs. \$725K in the West), suggesting untapped market opportunity. Conducting a South region market analysis to identify specific barriers (whether they are related to distribution, marketing presence, or competitive factors) would be a logical first step. Targeted initiatives to address identified gaps could help close the performance disparity.

4. Prioritize Technology Category Expansion

Technology generates 50.8% of total profit while representing only 36.4% of sales, with strong margins of 17.35%. This category represents the business's most profitable growth opportunity. Expanding the Technology product portfolio, increasing marketing investment in this category, and ensuring adequate inventory allocation could drive profitable revenue growth without the margin pressure seen in other categories.

5. Develop Customer Segmentation and Retention Strategy

The wide variation in customer profitability visible in the analysis suggests an opportunity for more strategic customer management. Identifying and focusing resources on the most profitable customers (through enhanced service, tailored pricing programs, or dedicated account management) could improve retention and increase customer lifetime value. Conversely, understanding which customer relationships are unprofitable would inform decisions about pricing adjustments or resource allocation.

FINAL REMARKS

This analysis reveals a company with strong growth momentum but critical profitability challenges. The good news: these challenges are addressable through systematic execution of the recommendations outlined above. The company has already demonstrated operational capability through its growth achievements, now it must channel that capability toward profitability optimization.

The path forward requires discipline (especially on discounting), strategic courage (particularly regarding Furniture), and sustained focus on profit quality over mere revenue growth. Success will transform the business from a fast-growing but marginally profitable operation into a high-performing, sustainably profitable enterprise.

The recommendations are deliberately sequenced to generate quick wins (discount governance, South region focus) that build momentum and fund longer-term investments (analytics, portfolio transformation). Execution discipline will be critical; these initiatives require sustained management attention and accountability.

The single most important principle: Every decision should be evaluated through the lens of profit contribution, not just revenue impact. Growth is important, but profitable growth is essential.